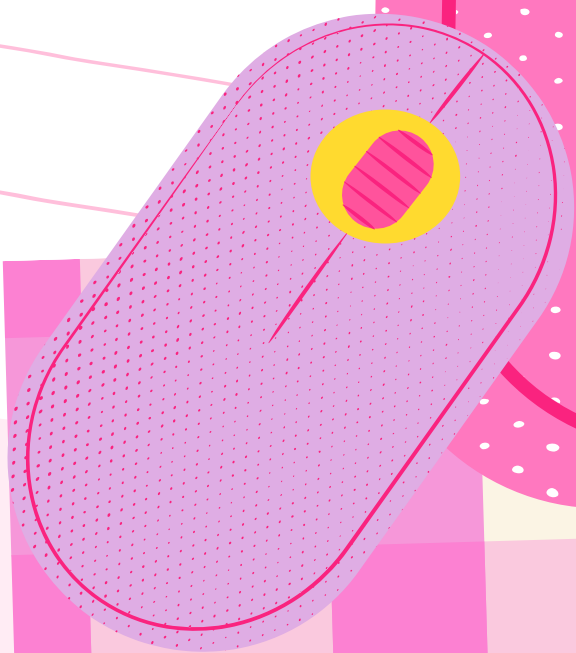
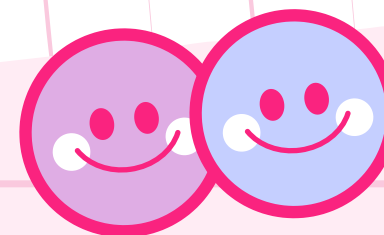
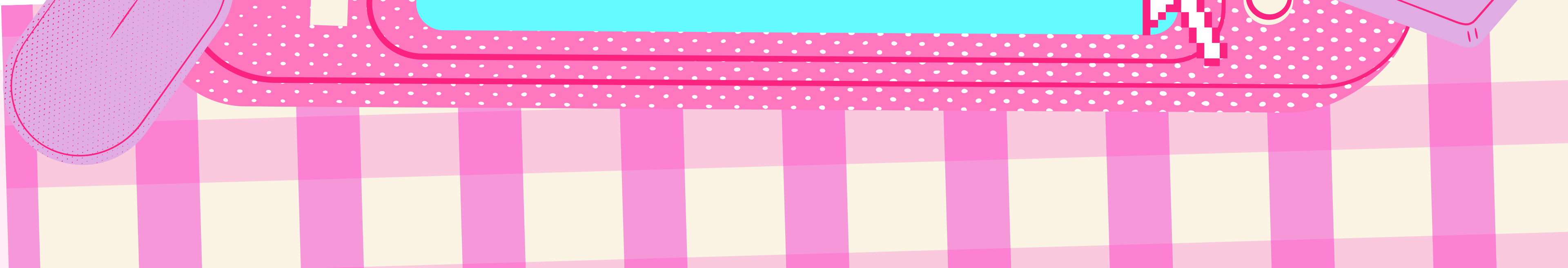
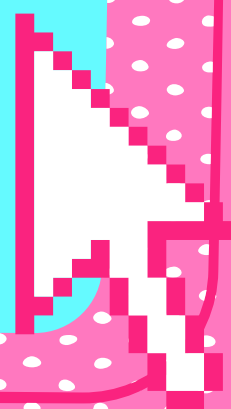


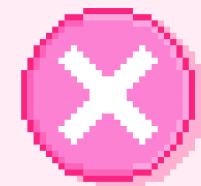


Runningwater
Remains

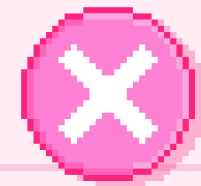




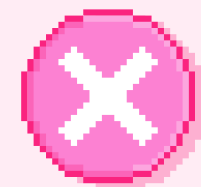
Anggota



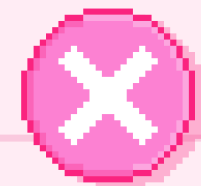
ATTHORIQ BAGUS FADILLAH
(25/568233/SV/27400)



HAIDAR DAFFA RASYIQIN
(25/562020/SV/26719)



ILHAM ANTARIKSA WIDODO
(25/561192/SV/26529)

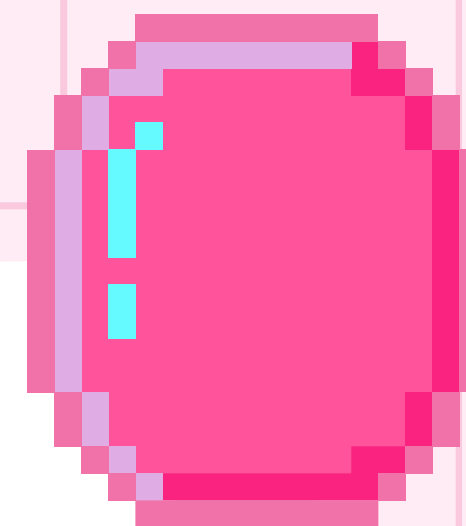
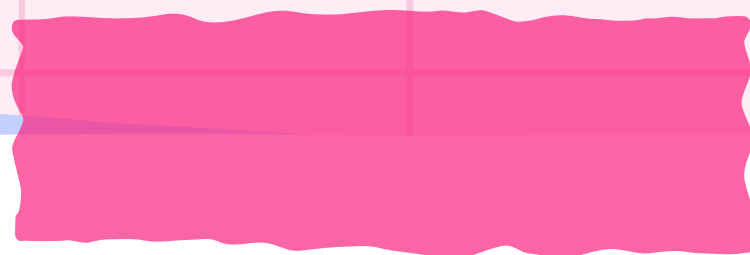
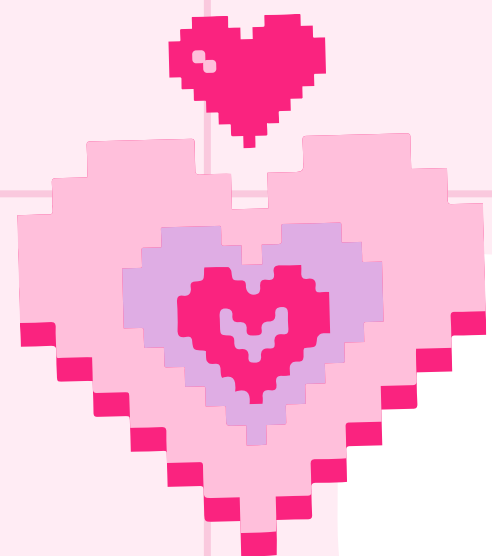


NIZAR ALDIN RASYADI
(25/568541/SV/27488)

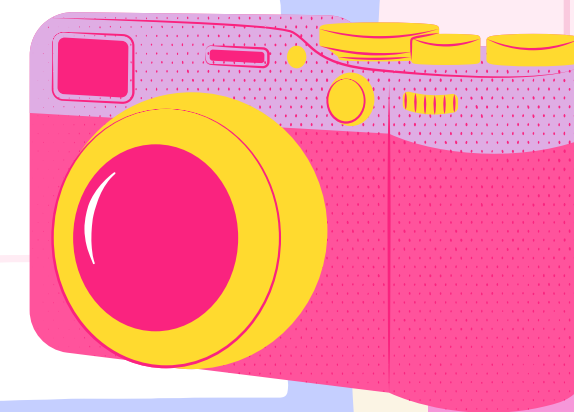


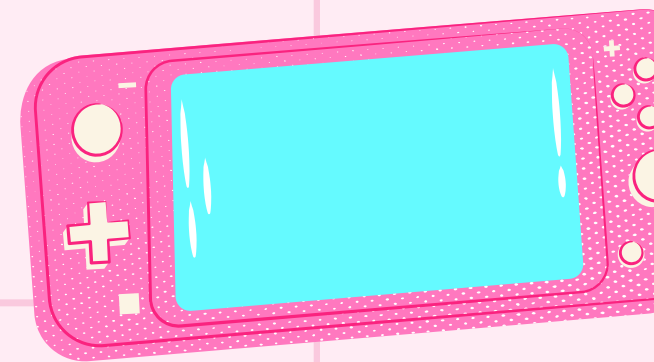
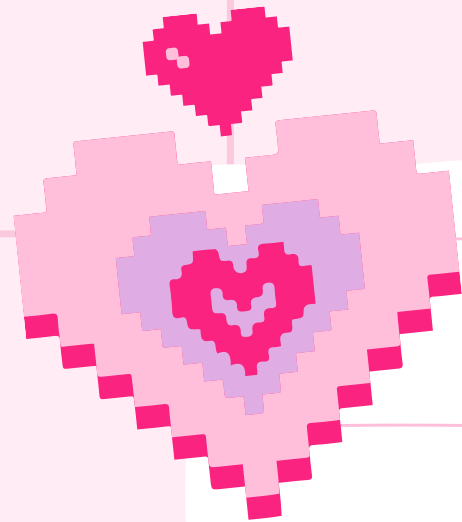
NYOMAN DIMAS WIRA BAKTI
(25/561741/SV/26643)





KONSEP GAME





RUNNINGWATER REMAINS

ROLE-PLAYING GAME (RPG)
BERBASIS GILIRAN (TURN-BASED)

Perkembangan karakter:

- Mengalahkan Musuh +40 XP dan +50 skor
- Naik level: maksimum HP bertambah 20 dan HP karakter pulih
- Setelah semua musuh mati: pemain mendapat 0–2 item loot secara acak

Kondisi menang: Mengalahkan seluruh musuh di Stage 3 → Victory

Fitur tambahan: Sistem simpan/muat (save/load), inventaris item (senjata, baju besi, tools, konsumabel), dan animasi sprite

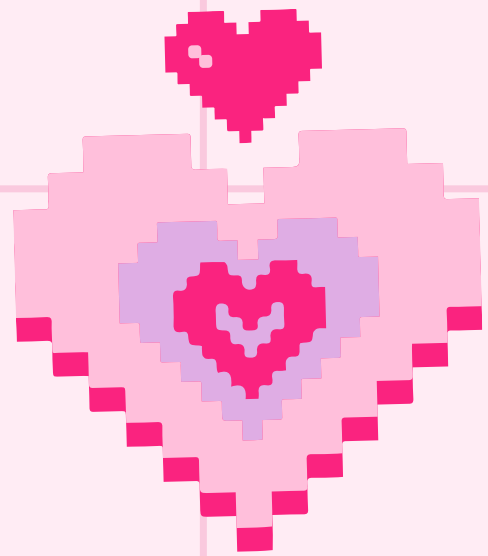
ATTACK (Z)

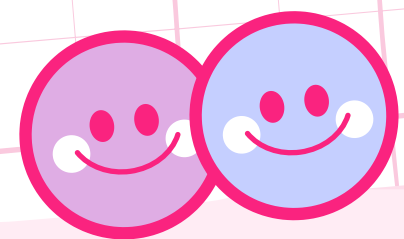
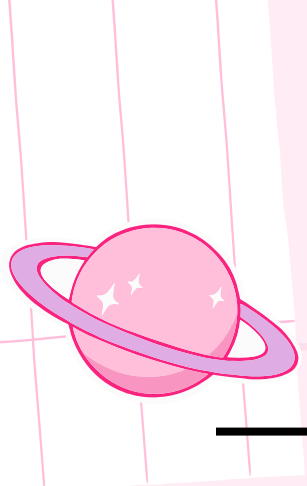
ABILITY (A)

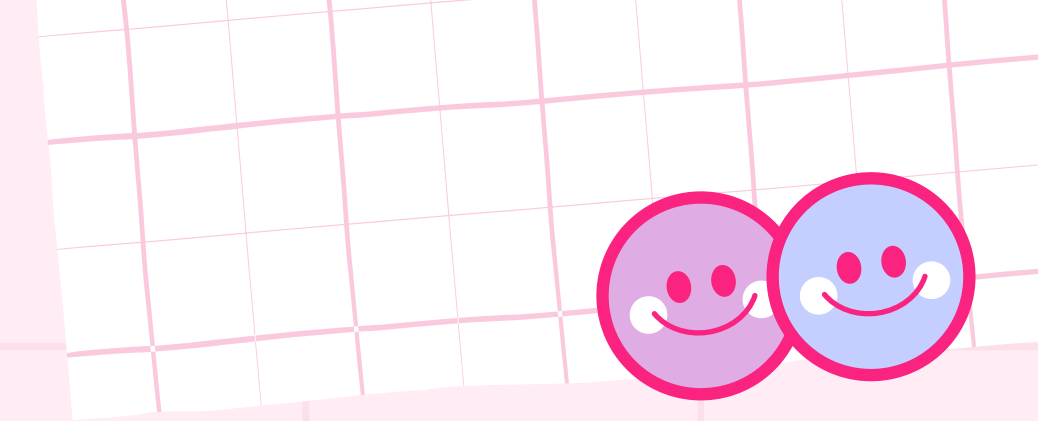
ITEM (X)

FLEE

TUJUAN



	
ENCAPSULATION	Seluruh field private/protected, akses lewat getter/setter
INHERITANCE	PlayChar & Enemies extends Entity; semua item extends Item
POLYMORPHISM	GameManager.Tick() memanggil UpdateSystem() pada semua sistem secara polimorfis
METHOD OVERLOADING	AddScore(int) vs AddScore(int, float) di ScoringSystem
 Total: 7 Konsep OOP	



METHOD OVERRIDING

TakeDamage() di PlayChar dan Enemies meng-override milik Entity

ABSTRACT CLASS

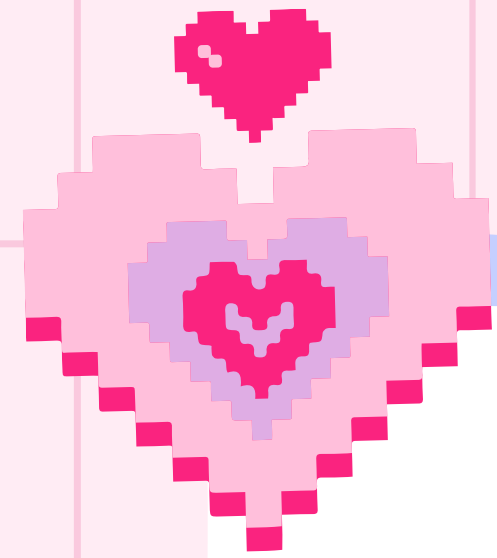
Entity, Item, dan GameSystem sebagai kelas abstrak fondasi

INTERFACE

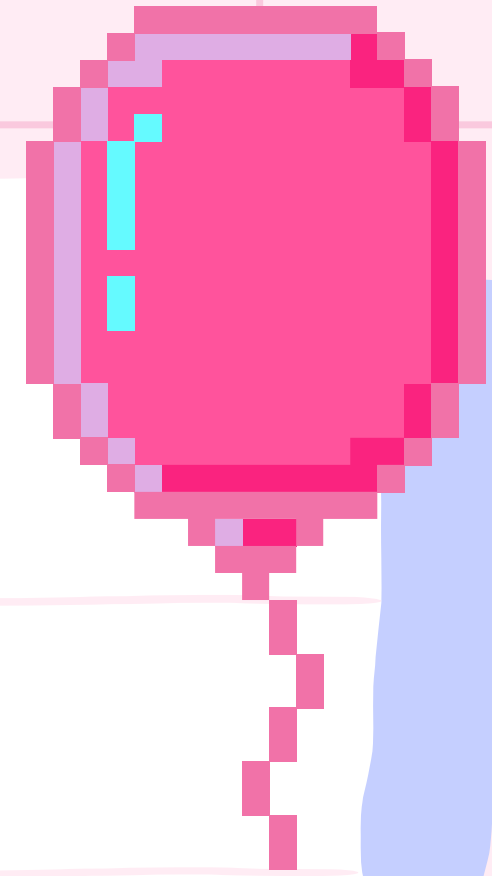
ISavable diimplementasikan oleh PlayChar, Stages, dan GameManager



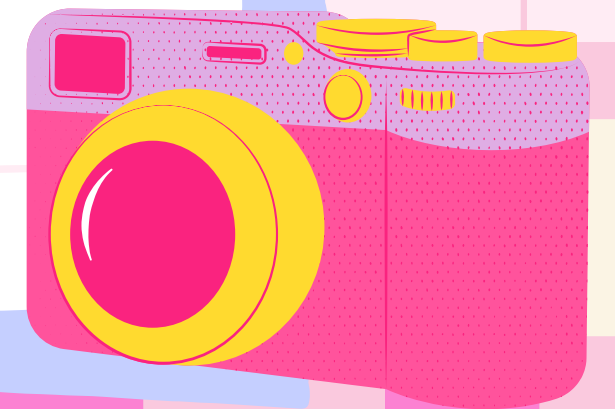
Total: 7 Konsep OOP

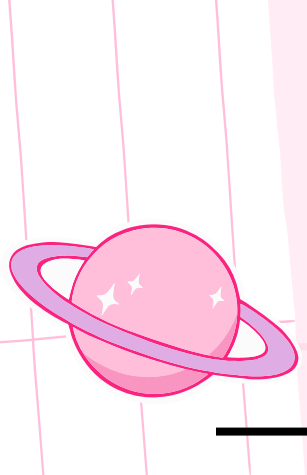


2.
Desain Sistem

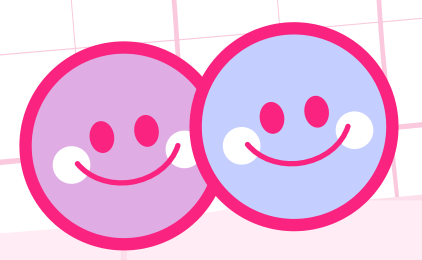


CLASS & UML DIAGRAM





ARSITEKTUR CLASS GAME



MANAJEMEN GAME

GameManager, Stages

HIERARKI ENTITY

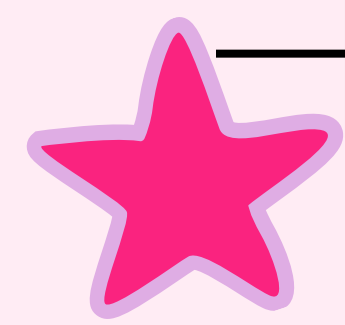
Entity (abstract),
PlayChar, Enemies

HIERARKI ITEM

Item (abstract), Weapon, Armour, Tools, Consumables

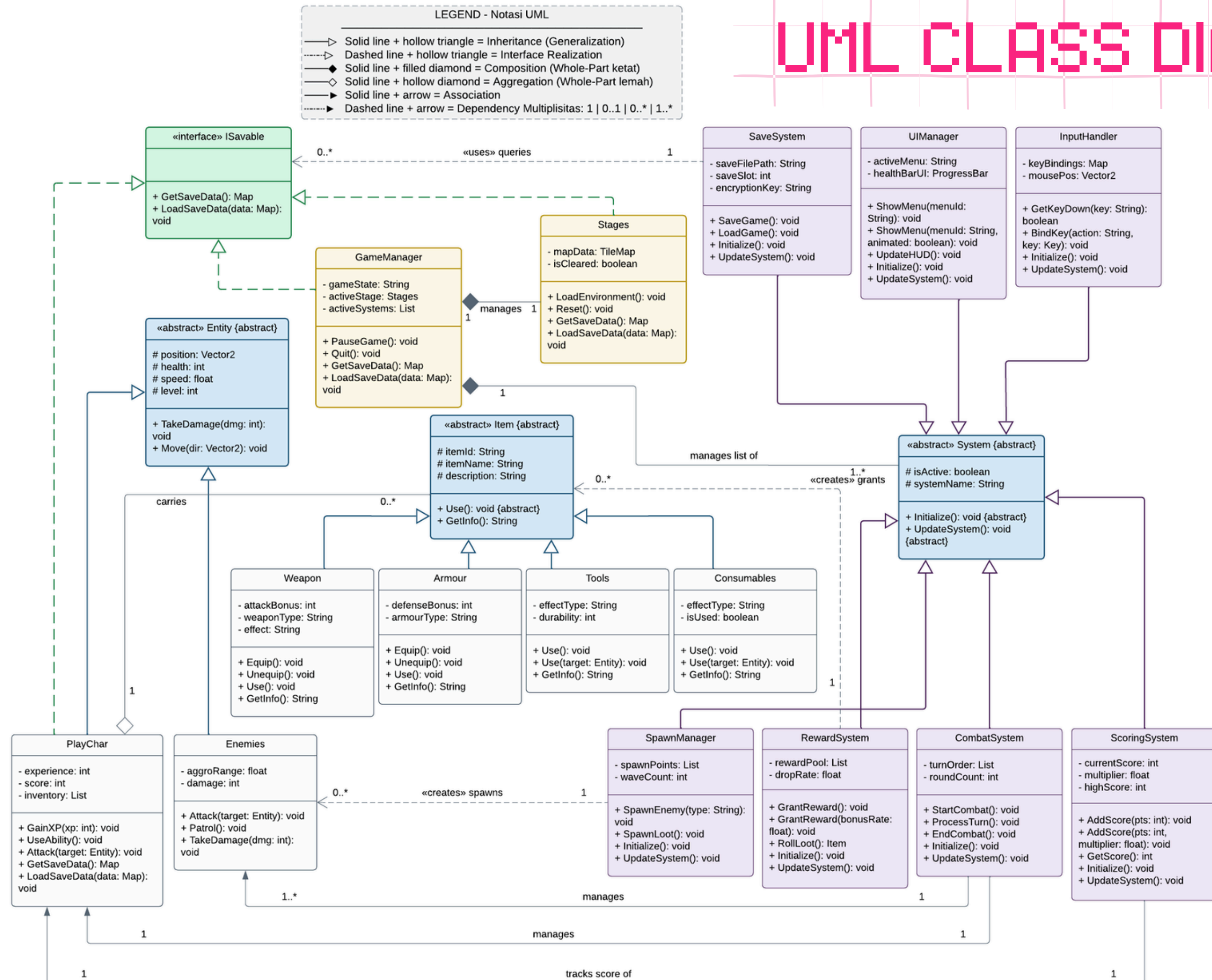
HIERARKI SYSTEM

GameSystem (abstract), + 7 manager



Total: 17 class konkret + 3 abstract class + 1 interface

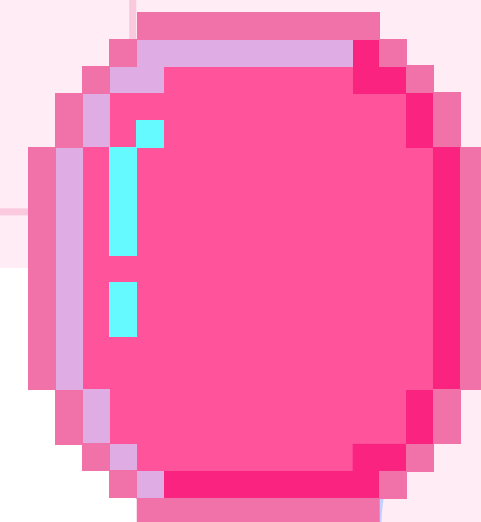
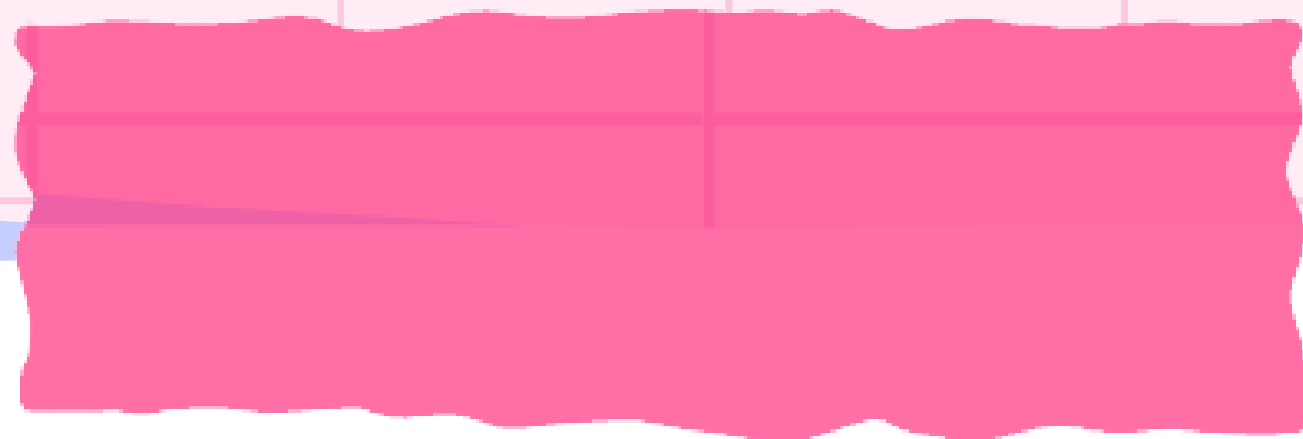
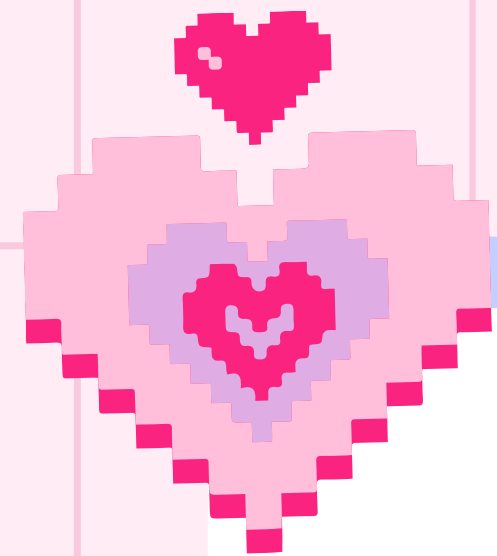
UML CLASS DIAGRAM



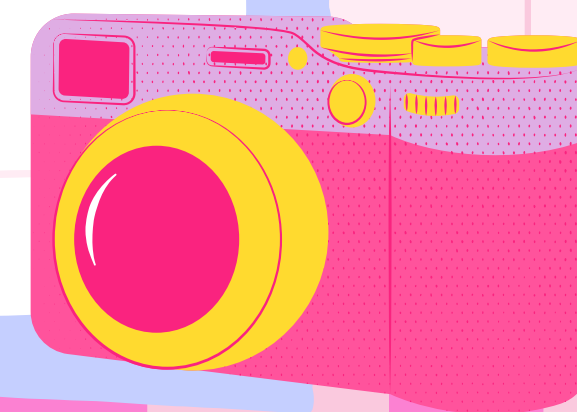
RELASI ANTAR CLASS

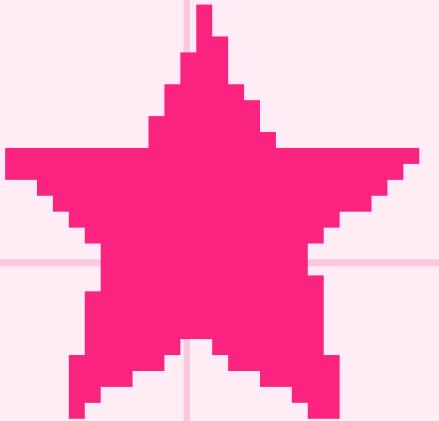
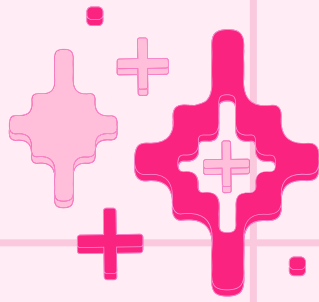
CLASS A	RELASI	CLASS B
PlayChar / Enemies	Inheritance	Entity
Weapon / Armour / Tools / Consumables	Inheritance	Item
PlayChar, Stages, GameManager	Realization (implements)	ISavable
GameManager	Composition ◆	Stages & GameSystem
PlayChar	Aggregation ◇	Item (inventory)
SaveSystem	Dependency	ISavable
CombatSystem	Association	PlayChar, Enemies

Relasi composition dipakai karena Stages & System tidak bisa eksis tanpa GameManager



IMPLEMENTASI OOP





ENCAPSULATION

CLASS

PRIVATE FIELD

AKSES

ScoringSystem

currentSystem, multiplier

AddScore(), GetScore()

PlayChar

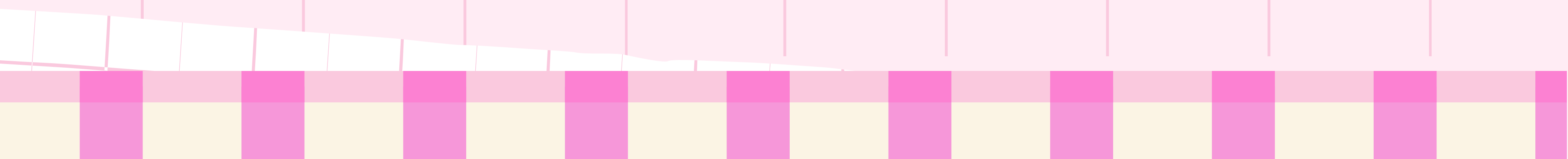
inventory, equippedWeapon

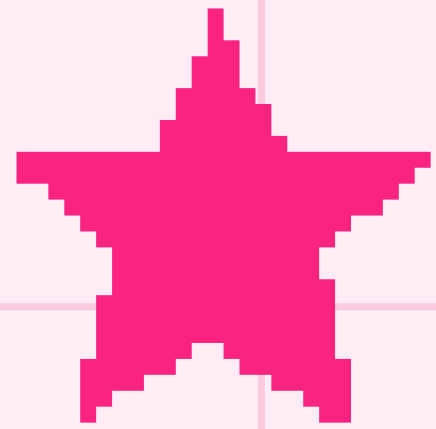
equipWeapon(),
Attack()

Enemies

damage, aggroRange

getDamage(), Attack()





INHERITANCE

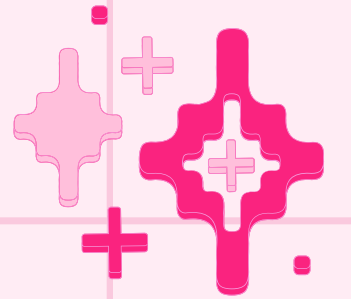
ENTITY

- PlayChar - GainXp, inventory
- Enemies - AIPatrol, level reduction

ITEM

- Weapon - AttackBonus, equip
- Armour - DefenseBonus, equip

POLYMORPHISM



GAME LOOP

- GameManager mengiterasi list activeSystems dan memanggil UpdateSystem() tanpa perlu tahu tipe riilnya.

COMBAT LOGIC

- Menangani target.TakeDamage() secara seragam baik untuk Player maupun Enemy.

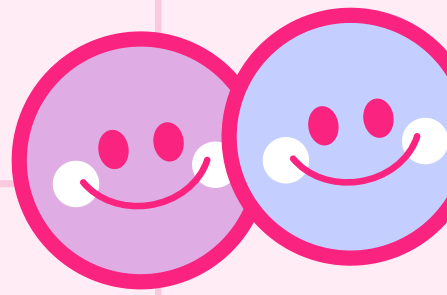


OVERLOADING

Beberapa method dengan nama sama namun parameter berbeda (Compile-time).

CONTOH

AddScore(int pts) pada ScoringSystem.

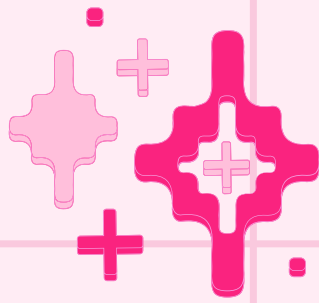


OVERRIDING

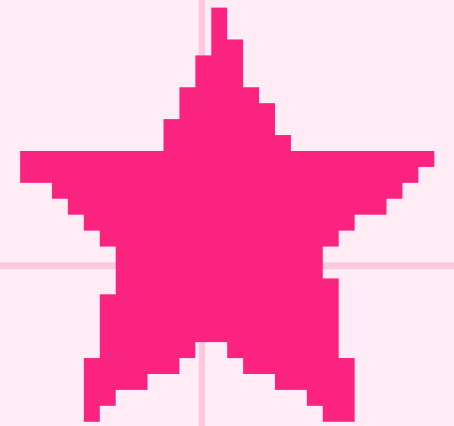
Menulis ulang implementasi method superclass di subclass (Runtime).

CONTOH

TakeDamage() di PlayChar menerapkan armor, sedangkan di Enemies menerapkan level reduction.



ABSTRACT CLASS



ENTITY

Definisi dasar objek dunia game

Abstract method:

➤ Move(Vector2 dir)

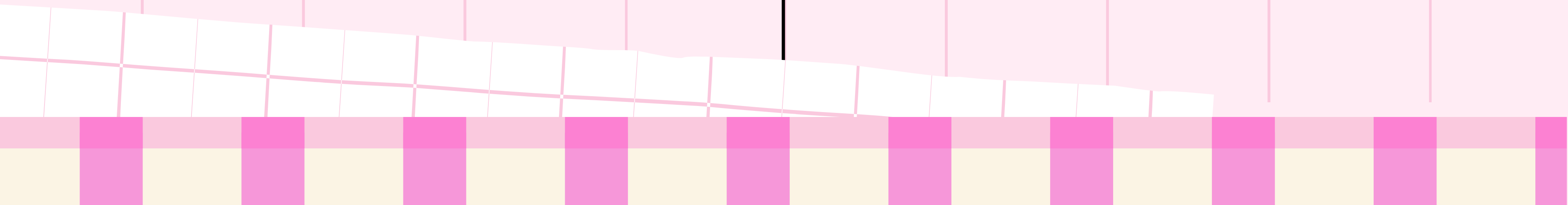
ITEM

Spesifikasi barang interaktif

Abstract method:

➤ Use()

➤ Use(Entity target)





INTERFACE

ISAVABLE

Digunakan sebagai kontrak untuk fitur Save & Load tanpa bergantung pada hierarki inheritance.

METHOD KONTRAK

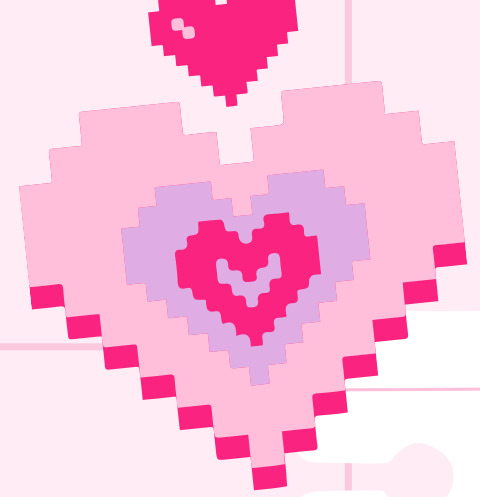
- `Map<String, Object> getSaveData()`
- `void loadSaveData(Map data)`

Diimplementasikan oleh PlayChar, Stages, dan GameManager.





i DEMO
GAME



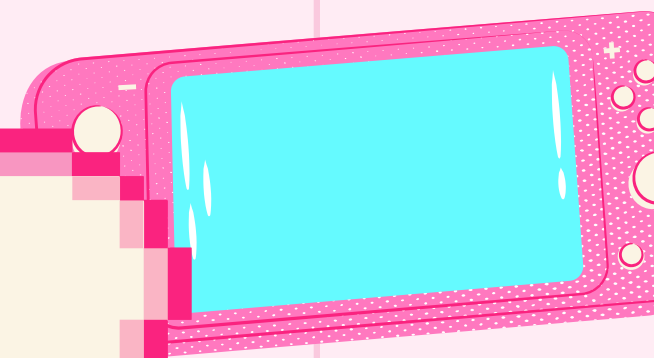
1. MODULAR & SCALABLE

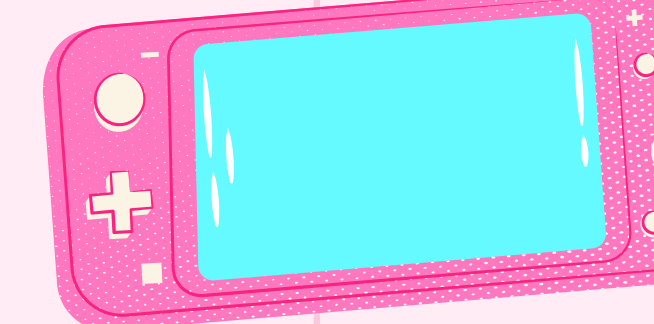
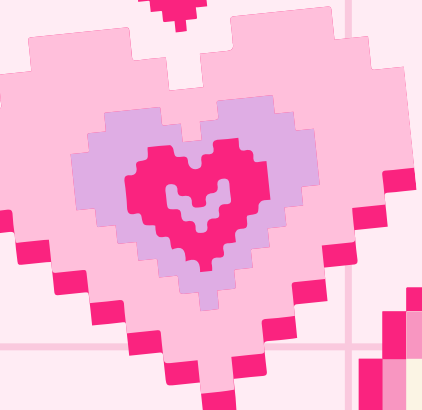
2. GAME LOOP ROBUST

3. PROGRESSION SYSTEM

KELEBIHAN

SYSTEM





KEKURANGAN

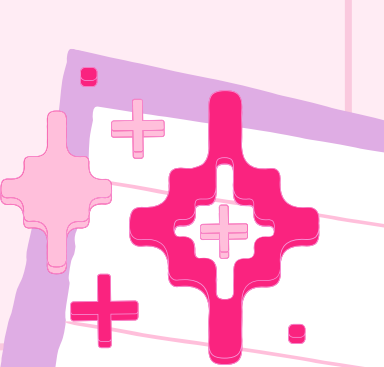
SYSTEM

1.KONTEN TERBATAS

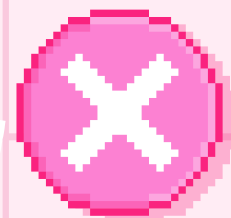
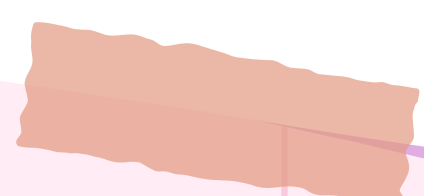
2.STRATEGI MINIMAL

3.AI MUSUH TERLALU
PREDICTABLE

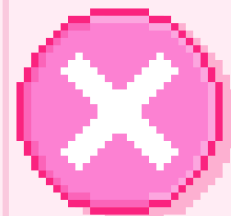




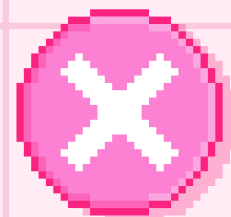
Future Plans



MEMPERKUAT AI MUSUH & MENAMBAHKAN VARIETY ENEMY



MENAMBAHKAN LEBIH BANYAK STAGE BESERTA DENGAN MEKANIK KHUSUS STAGE



IMPLEMENTASI NPC YANG DAPAT MEMBANTU

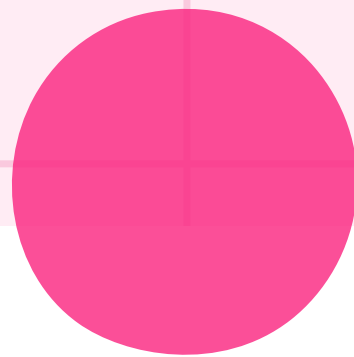
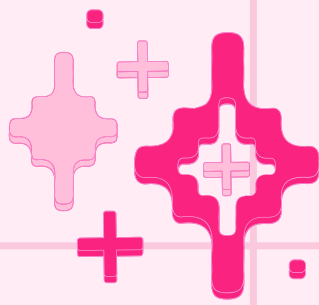


MENAMBAH JUMLAH SKILL YANG DAPAT DIGUNAKAN SERTA MEMBUAT SKILL TREE



MEMBUAT SISTEM UNTUK MEMBUAT MAP SECARA PROSEDURAL





THANK YOU

